Experiment 18: Fruit Color Classification

Aim:

To implement a production-rule based system that classifies fruits according to their color.

Algorithm:

1. Define production rules of the form IF color THEN fruit.
2. Accept color as input.
3. Fire matching rules.
4. Output the classified fruit(s).

Code:

```
def classify_fruit(color):
    rules = {
        'red': ['Apple', 'Strawberry', 'Cherry'],
        'yellow': ['Banana', 'Mango', 'Lemon'],
        'orange': ['Orange', 'Peach'],
        'green': ['Grapes', 'Apple', 'Pear'],
        'purple': ['Grapes', 'Plum']
    }
    return rules.get(color.lower(), ['Unknown'])

print("Red fruits:", classify_fruit('red'))
print("Yellow fruits:", classify_fruit('yellow'))
```

Sample Output:

```
Red fruits: ['Apple', 'Strawberry', 'Cherry']
Yellow fruits: ['Banana', 'Mango', 'Lemon']
```

Result:

The production rule system correctly classified fruits based on the given color attribute.